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Physical and Cognitive Development in Early Adulthood

CHAPTER OUTLINE

THE TRANSITION FROM ADOLESCENCE TO ADULTHOOD

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Achievement
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Work

Stories of Life-Span Development: Dave Eggers, Pursuing a Career in the Face of Stress

He was a senior in college when both of his parents died of cancer within five weeks of each other. What would he do? He and his 8-year-old brother left Chicago to live in California, where his older sister was entering law school. Dave would take care of his younger brother, but he needed a job. That first summer, he took a class in furniture painting; then he worked for a geological surveying company, re-creating maps on a computer. Soon, though, he did something very different: With friends from high school, Dave Eggers started *Might*, a satirical magazine for twenty-somethings. It was an edgy, highly acclaimed publication, but not a moneymaker. After a few years, Eggers had to shut down the magazine, and he abandoned California for New York.

This does not sound like a promising start for a career. But within a decade after his parents' death, Eggers had not



Dave Eggers is a talented and insightful author.
Andrea Sabbadini/Alamy Stock Photo

only raised his young brother but had also founded a quarterly journal and Web site, *McSweeney's*, and had written a best-seller, *A Heartbreaking Work of Staggering Genius*, which received the National Book Critics Circle Award and was nominated for a Pulitzer Prize. It is a slightly fictionalized account of Eggers' life as he helped care for his dying mother, raised his brother, and searched for his own place in the world. Despite the pain of his loss and the challenges involved in raising his brother, Eggers quickly built a record of achievement as a young adult. ■

emerging adulthood A period of transition from adolescence to adulthood (approximately 18 to 25 years of age) that involves experimentation and exploration.

The Transition from Adolescence to Adulthood

When does an adolescent become an adult? It is not easy to tell when a girl or a boy enters adolescence. The task of determining when an individual becomes an adult is even more difficult.

Becoming an Adult

For most individuals, becoming an adult involves a lengthy transition period (Nice & others, 2023). The transition from adolescence to adulthood has been referred to as **emerging adulthood**, a life stage that lasts from approximately 18 to 25 years of age (Arnett, 2006, 2015). Experimentation and exploration characterize the emerging adult. At this point in their development, many individuals are still exploring which career path they want to follow, what they want their identity to be, and which lifestyle they want to adopt (for example, being single, cohabiting, or getting married) (Kuang & others, 2023).

Key Features of Emerging Adulthood

Jeffrey Arnett (2006) has concluded that five key features characterize emerging adulthood:

- *Identity exploration, especially in love and work.* Emerging adulthood is the time during which key changes in identity take place for many individuals (McLean & others, 2021).
- *Instability.* Residential changes peak during early adulthood, a time during which there also is often instability in love, work, and education.
- *Self-focused.* According to Arnett (2006, p. 10), emerging adults “are self-focused in the sense that they have little in the way of social obligations, little in the way of duties and commitments to others, which leaves them with a great deal of autonomy in running their own lives.”

How Would You...?

As a human development and family studies professional, what characteristics of emerging adulthood might you share with college counselors?

- *Feeling in-between.* Many emerging adults don't consider themselves adolescents or full-fledged adults (Bellingtier & Neupert, 2020).
- *The age of possibilities, a time when individuals have an opportunity to transform their lives.* Arnett (2006) describes two ways in which emerging adulthood is the age of possibilities: (1) Many emerging adults are optimistic about their future; and (2) for emerging adults who have experienced difficult times while growing up, emerging adulthood presents an opportunity to reorient their lives in a more positive direction.



What characterizes emerging adults?
Peter Code/Photodisc/Getty Images



Research indicates that these five aspects characterize not only individuals in the United States as they make the transition from adolescence to early adulthood, but also their counterparts in European countries and Australia (Arnett, 2015; Buhl & Lanz, 2007; Sirsch & others, 2009). Although emerging adulthood is not recognized as a developmental period in all cultures, it does appear to occur in those where assuming adult roles and responsibilities is postponed (Kins & Beyers, 2010). Critics of the concept of emerging adulthood argue that it applies mainly to privileged adolescents and is not always a self-determined choice for many young people, especially those in limiting socioeconomic conditions (Cote & Bynner, 2008). One study revealed that U.S. at-risk youth entered emerging adulthood slightly earlier than the general population of youth (Lisha & others, 2012).

Increasingly, instead of describing emerging adulthood as occurring from 18 to 25, the age range has been expanded to 18 to 29 (Arnett, 2014). However, in a recent study of individuals 18 to 60, the five characteristics of emerging adulthood mentioned above (identity exploration, self-focus, feeling in-between, instability, and possibilities/optimism) were more likely to be endorsed by 18- to 25-year-olds than by older individuals (Arnett & Mitra, 2020).

The Changing Landscape of Emerging and Early Adulthood

In earlier generations, by their mid-twenties at the latest, individuals were expected to have finished college, obtained a full-time job, and established their own household, most often with a spouse and a child. However, individuals are now taking much longer to reach these developmental milestones, many of which they are not experiencing until their late twenties or even thirties (Vespa, 2017). It is not surprising that their parents recall having had a much earlier timetable of reaching these developmental milestones.

Consider that for the first time in the modern era, in 2020 a majority (52 percent) of 18- to 29-year-olds lived with their parents for the first time since the Great Depression (Fry, Passel, & Cohn, 2020). Dating all the way back to 1880, living with a romantic partner, whether a spouse or a significant other, was previously the most common living arrangement for emerging and young adults. The increased proportion of post-adolescents living with their parents is most pronounced among 18- to 24-year-olds (71 percent in 2020). In terms of education, today's emerging and young adults are better educated than their counterparts in the 1970s (Vespa, 2017). For example, they are much more likely to hold a college degree today. The biggest reason for this educational improvement since the 1970s, though, is a gender difference reversal. In 1975, more young men had college degrees, but today more young women than young men have a college degree.

In terms of work, more young adults are working today than in 1975 (Vespa, 2017). The main reason for this increase also involves a gender change—the significant rise of young women in the workforce, which has increased from slightly below 50 percent to more than two-thirds of young women in the workforce today. In 1975, almost all of the women who were not in the workforce stated that they were taking care of their home and children. However, today, fewer than 50 percent of the women who were not in the workforce were homemakers.

We will further discuss these lifestyle changes in the chapter on “Socioemotional Development in Early Adulthood.”

Markers of Becoming an Adult

In the United States, the most widely recognized marker of entry into adulthood is holding a more or less permanent, full-time job, which usually happens when an individual finishes school—high school for some, college for others, graduate or professional school for still others. However, other criteria are far from clear. Economic independence is one marker of adult status, but achieving it is often a long process. College graduates are increasingly returning to live with their parents as they attempt to establish themselves economically. A longitudinal study found that at age 25 only slightly more than half of the participants were fully financially independent of their family of

origin (Cohen & others, 2003). The most dramatic findings in this study, though, involved the extensive variability in the individual trajectories of adult roles from 17 to 27 years of age; many of the study's participants moved back and forth between increasing and decreasing economic dependency. One study revealed that continued co-residence with parents during emerging adulthood slowed the process of becoming a self-sufficient and independent adult (Kins & Beyers, 2010).

Other studies suggest that taking responsibility for oneself is an important marker of adult status for many individuals (Smith & others, 2017). In one study, college students and their parents agreed that taking responsibility for one's actions and developing emotional control are important aspects of becoming an adult (Nelson & others, 2007). And in a study of Danish emerging adults, the most widely described markers of emerging adulthood were accepting self-responsibility, making independent decisions, and becoming financially independent (Arnett & Padilla-Walker, 2015). In this study the least-described markers were the traditional transition events of getting married and avoiding getting drunk. What we have discussed about the markers of adult status mainly characterizes individuals in industrialized societies, especially Americans. In developing countries, marriage is more often a significant marker for entry into adulthood, and this usually occurs much earlier than the adulthood markers in the United States (Arnett, 2015). In one study, the majority of 18- to 26-year-olds in India felt that they had achieved adulthood (Seiter & Nelson, 2010).

The Transition from High School to College

For many individuals in developed countries, entering college is an important aspect of the transition to adulthood (Stolzenberg & others, 2020). Just as the transition from elementary school to middle or junior high school involves change and possible stress, so does the transition from high school to college. The two transitions have many parallels. Going from being a senior in high school to being a freshman in college replays the top-dog phenomenon of transferring from the oldest and most powerful group of

students to the youngest and least powerful group of students that occurred as adolescence began. For many students, the transition from high school to college involves movement to a larger, more impersonal school structure; interaction with peers from more diverse geographical and sometimes more diverse ethnic backgrounds; and increased focus on achievement and its assessment. And like the transition from elementary to middle or junior high school, the transition from high school to college can involve positive features. College students are more likely to feel grown up, have more subjects from which to select, have more time to spend with peers, have more opportunities to explore different lifestyles and values, enjoy greater independence from parental monitoring, and be challenged intellectually by academic work (Halonen & Santrock, 2013).

Over the past three decades, the Higher Education Research Institute at UCLA has surveyed first-year college students' backgrounds, experiences, and views on a number of topics. In recent years, traditional-aged college students have shown an increased concern for personal well-being and a decreased concern for the well-being of others, especially for the disadvantaged (Stolzenberg & others,



The transition from high school to college often involves positive as well as negative features. In college, students are likely to feel grown up, able to spend more time with peers, have more opportunities to explore different lifestyles and values, and enjoy greater freedom from parental monitoring. However, college involves a larger, more impersonal school structure and an increased focus on achievement and its assessment. *What was your transition to college like?*
Stockbyte/Stockdisc/Getty Images

2020). Today's college freshmen are more strongly motivated to be well-off financially and less motivated to develop a meaningful philosophy of life than were their counterparts of 40 years ago. In 2019, 84.3 percent of students (the highest percent ever in this survey) viewed becoming well-off financially as an "essential" or a "very important" objective, compared with only 42 percent in 1971.

There are, however, some signs that U.S. college students are shifting toward a stronger interest in the welfare of society. In the survey just described, U.S. college freshmen's interest in developing a meaningful philosophy of life increased from 39 percent in 2001 to 49.8 percent in 2019 (Stolzenberg & others, 2020). In addition, in 2019, 38.2 percent of college freshmen said that they had frequently or occasionally engaged in volunteer work in the past year, a significant increase from 18 percent in 1990 (Stolzenberg & others, 2020)

An increasing number of first-year college students also report having high levels of stress and depression. In the national survey just described, in 2019, 42.7 percent (an increase from 38.7 percent in 2017) of first-year college students said they frequently or occasionally felt overwhelmed with all they had to do, 16.2 percent (an increase from 12.2 percent in 2017) indicated they were depressed, and 37.6 percent (an increase from 32.5 percent in 2017) reported feeling anxious (Stolzenberg & others, 2020).

In mid-March 2020, a large majority of colleges shut down in-person classes and abruptly switched to online instruction, leading to concerns that learning would be less effective and the social aspects of college would be considerably compromised. One study found that more than half of college students reported high levels of stress related to online classes they had to take during the COVID-19 pandemic (Sorgo & others, 2022). They reported that this stress involved their lack of motivation to study, poor quality of Internet and mobile phone connections, and presence of distracting factors.

Before the fall semester of 2020, college administrations grappled with whether to continue most or all classes online, return to in-person classes, or implement a hybrid schedule with some classes online and others in-person. A number of colleges that chose to have in-person classes had to return to online classes because of spikes in the spread of the coronavirus, especially fueled by college students not adequately engaging in preventive measures such as social distancing, mask wearing, and avoiding large gatherings.

Traditional-age college students in their late teens and early twenties are less likely to get severe symptoms from the coronavirus, but for older college students and faculty the risk is much greater. International students also are vulnerable, with travel bans to and from the United States compounding the difficulties of campus closings as well as increasing concern about housing options. Further, many students from low-income backgrounds may not have adequate access to computers and online instruction.

As the COVID-19 pandemic began subsiding in late 2021, most colleges were returning to a menu of mainly in-person classes or offering both in-person and online classes. One of the very few positive outcomes of the pandemic was motivating colleges to improve the quality of their online instruction.



How has the COVID-19 pandemic influenced the way college students experience their education?
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Physical Development

As emerging and young adults learn more about healthy lifestyles and how they contribute to a longer life span, they are increasingly interested in monitoring their physical performance, health, nutrition, exercise, and substance use.

Physical Performance and Development

Most of us reach our peak physical performance before the age of 30, often between the ages of 19 and 26. This peak of physical performance occurs not only for the

average young adult but for outstanding athletes as well. Even though athletes as a group keep getting better than their predecessors—running faster, jumping higher, and lifting more weight—the age at which they reach their peak performance has remained virtually unchanged.

Different types of athletes, however, reach their peak performances at different ages. Most swimmers and gymnasts peak in their late teens. Golfers and marathon runners tend to peak in their late twenties. In other areas of athletics, peak performance often occurs in the early to mid-twenties.

Not only do we reach our peak physical performance during early adulthood, it is also during this age period that we begin to decline in physical performance. Muscle tone and strength usually begin to show signs of decline around the age of 30. Sagging chins and protruding abdomens also may begin to appear for the first time. The lessening of physical abilities is a common complaint among the just-turned thirties. Sensory systems show little change in early adulthood, but the lens of the eye loses some of its elasticity and becomes less able to change shape and focus on near objects. Hearing peaks in adolescence, remains constant in the first part of early adulthood, and then begins to decline in the last part of early adulthood. And in the middle to late twenties, the body's fatty tissue increases.

Health

Emerging adults have more than twice the mortality rate of adolescents (Park & others, 2006). As indicated in Figure 1, males are mainly responsible for the higher mortality rate of emerging adults.

Although emerging adults have a higher death rate than adolescents, emerging adults have few chronic health problems, and they have fewer colds and respiratory problems than they did when they were children (Rimsza & Kirk, 2005). Although most college students know how to prevent illness and promote health, they don't fare very well when it comes to applying this information to themselves (Lau & others, 2013). In many cases, emerging adults are not as healthy as they seem to be (Fatusi & Hindin, 2010).

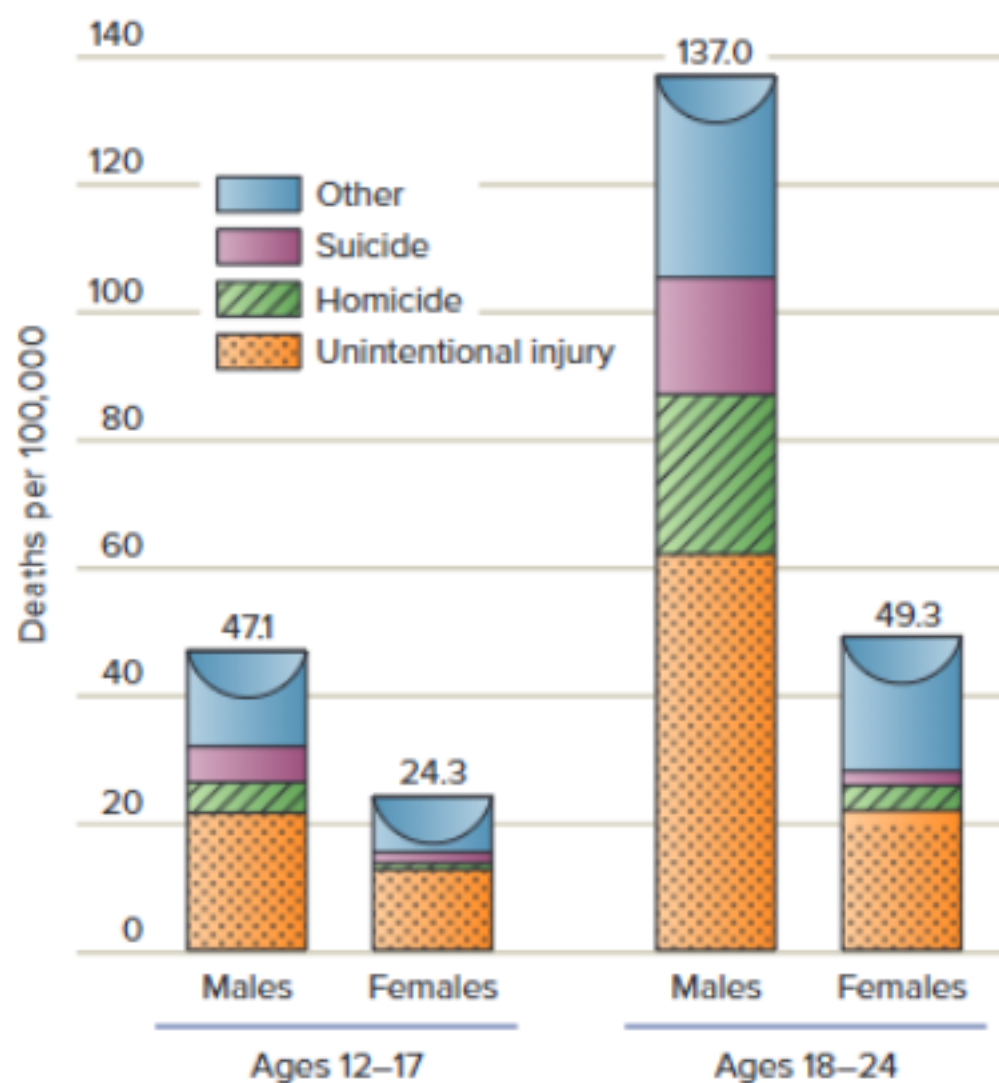


Figure 1 Mortality Rates of U.S. Adolescents and Emerging Adults.

adult lives. Many emerging adults develop a pattern of not eating breakfast, not eating regular meals, relying on snacks as their main food source during the day, eating excessively to the point where they exceed the normal weight for their age, smoking

A longitudinal study revealed that most bad health habits that were engaged in during adolescence increased in emerging adulthood (Harris & others, 2006). Inactivity, poor food choices, obesity, substance use, lack of reproductive health care, and insufficient health-care access worsened in emerging adulthood. For example, at 12 to 18 years of age, only 5 percent reported no weekly exercise, but at 19 to 26 years of age, 46 percent said they did not exercise during a typical week. And another study found that rates of being overweight or obese increased from 25.6 percent for college freshmen to 32 percent for college seniors (Nicoteri & Miskovsky, 2014). However, as with any period of development, there are clusters of emerging adults who are engaging in health-enhancing behaviors while others are engaging in health-damaging behaviors (Achak & others, 2023).

In emerging and early adulthood, few individuals stop to think about how their personal lifestyles will affect their health later in their

moderately or excessively, drinking moderately or excessively, failing to exercise, getting only a few hours of sleep at night, and engaging in risky sexual behavior (Andraka-Christou & others, 2020; Jansen & others, 2021). And a recent study of college students revealed that regularly eating breakfast was positively linked to their grade point average (GPA), while regularly eating fast food was negatively associated with their GPA (Reuter, Forster, & Brister, 2021).

Research indicates that 70 percent of college students do not get adequate sleep and that 50 percent report daytime sleepiness (Hershner & Chervin, 2015). A recent recommendation from the National Sleep Foundation (2023) is that emerging and young adults should get 7 to 9 hours of sleep a night, although many in these age groups do not get this much sleep (Nicholson & others, 2023). In a recent study of college students, sleep deprivation was linked to having a lower GPA and delayed college graduation (Chen & Chen, 2019). Also, in a recent study involving female college students, poorer sleep quality was associated with smartphone dependence and less effective stress management (Wang & others, 2019). In addition, research indicated that higher consumption of energy drinks was linked to more sleep problems in college students (Faris & others, 2017). A statement by the American Academy of Sleep Medicine and Sleep Research Society (Luyster & others, 2012) emphasized that chronic sleep deprivation may contribute to cardiovascular disease, a shortened life span, and impairment of cognitive and motor skills that increases the risk of motor vehicle crashes and work-related accidents.

Many college students occasionally or often pull an all-nighter to cram for an exam. Recently, the National Sleep Foundation (Suni & Rehman, 2022) described why this could harm your exam performance: It can distort your memory, cause thinking to be less clear, and impair concentration. Thus, your exam performance will likely be much better if you engage in effective planning and time management and spread your studying over a number of days and weeks.

The lifestyles just described are associated with poor health, which in turn reduces life satisfaction (Insel & Roth, 2024). In the Berkeley Longitudinal Study, in which individuals were evaluated over a period of 40 years, physical health at age 30 predicted life satisfaction at age 70, more so for men than for women (Mussen, Honzik, & Eichorn, 1982). Also, a recent study found that after the onset of the COVID-19 pandemic, college students' externalizing problems and attention problems increased (Copeland & others, 2021).

Eating and Weight

Obesity is a serious and pervasive health problem for many individuals (Fahey & others, 2023). In the United States, 42.4 percent of adults 20 to 39 years of age were classified as obese in 2020 (Centers for Disease Control and Prevention, 2022). This represents an 11 percent increase since 2000 and a 7 percent increase since 2012. There was no significant difference in the obesity rates of 20- to 39-year-olds in 2020 compared with 2018 (Hales & others, 2020). In terms of ethnic groups, in 2020, among adults 20 years of age and older, Black Americans and Latino adults had the highest obesity rates (49.9 and 45.6 percent, respectively), followed by non-Latino White adults (41.4 percent). Asian American adults had the lowest rate of obesity by far (16.1 percent). The highest rate of obesity (combining gender and ethnicity) occurred in Black American women (56.9 percent), the lowest in Asian American women (17.3 percent). In a recent international comparison of 33 countries, in 2020, Mexico had the highest overweight or obesity rate (75 percent), followed by the United States (73 percent); Japan had the lowest rate (27 percent), followed by South Korea (34 percent) (OECD, 2022).

Being overweight or obese is linked to increased risks of hypertension, diabetes, and cardiovascular disease (Teague & others, 2024). Overweight and obesity also are associated with mental health problems such as depression (Corbin & others, 2023).

How Would You...?

As a health-care professional, what research-based findings might you use to argue for the benefits of weight-reduction programs in colleges and community centers?



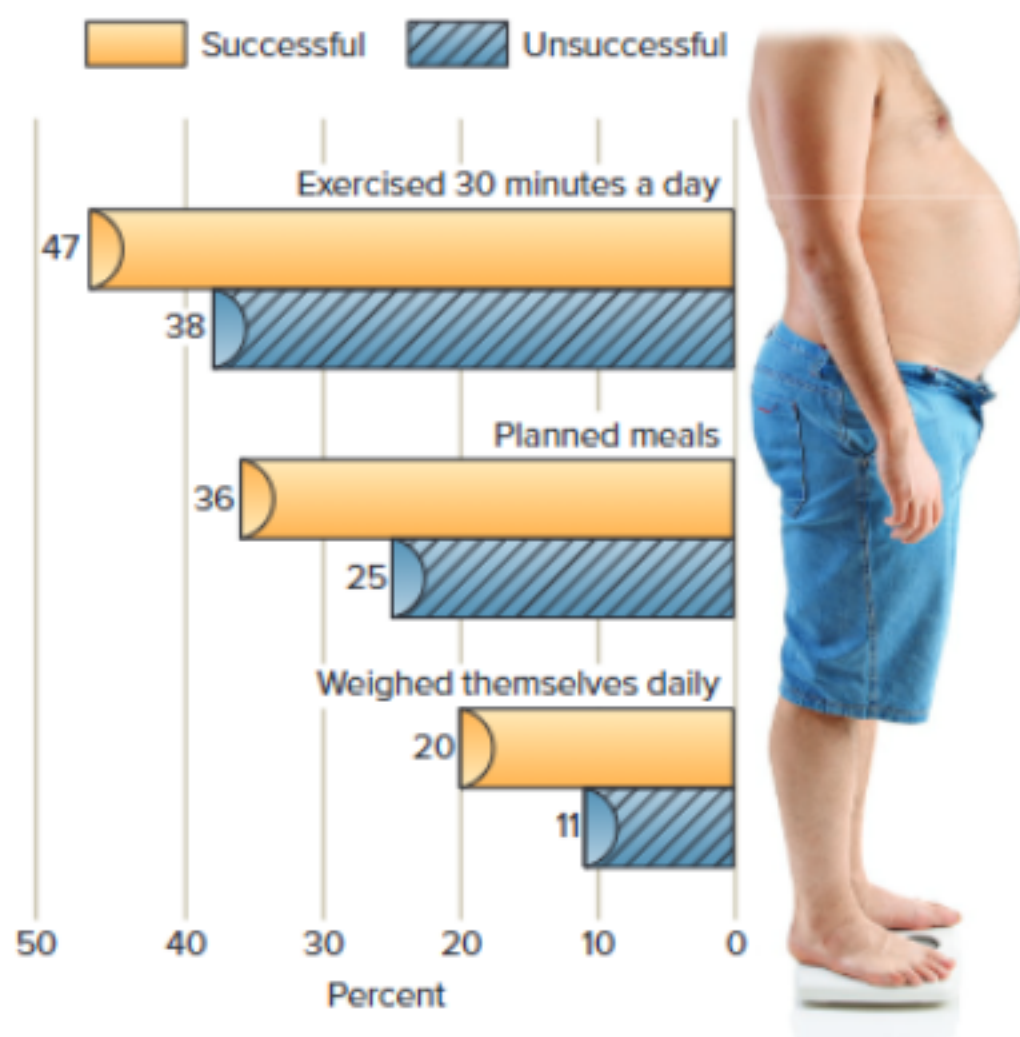


Figure 2
Comparison of
Strategies Used
by Successful and
Unsuccessful
Dieters.
Ljupco Smokovski/
Essentials/Getty Images

Also, in a recent study, women, especially those who were overweight, reported increased anxiety, depression, and loneliness during the COVID-19 pandemic (Lofrano-Prado & others, 2021).

One thing we know about losing weight is that the most effective programs include exercise (Annesi & Stewart, 2023). Researchers have found that higher levels of physical activity, especially endurance training, are linked to weight loss maintenance (Powers & others, 2024). Also, a study of approximately 2,000 U.S. adults found that exercising 30 minutes daily, planning meals, and weighing themselves daily were the strategies used more often by successful dieters than by unsuccessful dieters (Kruger, Blanck, & Gillespie, 2006) (see Figure 2).

In addition, a longitudinal study that followed individuals from adolescence through early adulthood found that the likelihood of maintaining a healthy weight was linked to having higher body satisfaction, avoiding unhealthy weight control behaviors such as skipping meals, and not engaging in highly restrictive dieting (Larson & others, 2018). Also in this study, social support for healthy eating and physical activity were linked to healthy weight maintenance, while having close relationships with individuals who were dieting (parents and significant others, for example) reduced the likelihood of maintaining a healthy weight.

Regular Exercise

One of the main reasons health experts want people to exercise is that it helps to prevent disorders such as heart disease and diabetes (Wuest & Walton-Fisette, 2024). Many health experts recommend that young adults engage in 30 minutes or more of aerobic exercise daily. **Aerobic exercise** is sustained exercise—jogging, swimming, or cycling, for example—that stimulates heart and lung activity. Most health experts recommend exercising vigorously enough to reach 60 percent of one's maximum heart rate. Only about one-fifth of adults, however, meet these recommended levels of physical activity.

The World Health Organization (2023) recognizes physical inactivity as a key factor in dying earlier. A national U.S. poll conducted by the Centers for Disease Control and Prevention found that only 22.9 percent of U.S. adults 18 to 64 years of age met federal guidelines for aerobic and muscle-strengthening exercise (Blackwell & Clarke, 2018). In this poll, 27.2 percent of men and 18.7 percent of women met the exercise guidelines.

Researchers have found that exercise improves physical and mental health (Powers & others, 2024). In particular, exercise is associated with a higher self-concept, as well as lower rates of anxiety and depression (Brush & others, 2022). For example, in a recent study of almost 18,000 adults, engaging in both regular moderate-to-vigorous physical activity and muscle-strengthening exercise was associated with the lowest incidence of depressive symptoms (Bennie & others, 2020). Also, on days when 18- to 25-year-old college students engaged in more physical activity, they reported greater satisfaction with life (Maher & others, 2013). Another study found a reduction in mortality risk when screen time was replaced by increased physical activity (Wijndaele & others, 2017).

Substance Abuse

Earlier, we explored substance abuse in adolescence. In one study, only 20 percent of college students reported abstaining from alcohol (Huang &

aerobic exercise Sustained exercise (such as jogging, swimming, or cycling) that stimulates heart and lung activity.

others, 2009). Fortunately, by the time individuals reach their mid-twenties, many have reduced their use of alcohol and drugs (Jager & others, 2023). As in adolescence, college and young adult men are more likely to take drugs than college and young adult women are (Patrick & others, 2022).

Heavy binge drinking often occurs in college, and it can take its toll on students (Jager & others, 2023). In 2021, 30 percent of U.S. college students reported having had five or more drinks in a row at least once in the last two weeks (Patrick & others, 2022). The term *extreme binge drinking* (also called high-intensity drinking) describes individuals who had 10 or more drinks in a row (Patrick & others, 2022). In 2021, 10.5 percent of college students reported drinking this heavily (Patrick & others, 2022). While drinking rates among college students have remained high, drinking, including binge drinking, has declined in recent years. For example, binge drinking declined from 37 percent in 2012 to 30 percent in 2018 (Patrick & others, 2022).

In a national survey of drinking patterns on 140 campuses (Wechsler & others, 1994), almost half of the binge drinkers reported problems that included missing classes, sustaining physical injuries, experiencing troubles with police, and having unprotected sex. For example, binge-drinking college students were 11 times more likely to fall behind in school, 10 times more likely to drive after drinking, and twice as likely to have unprotected sex in comparison with college students who did not binge drink. Further, a recent Korean study found that binge-drinking college students had deficits in memory and executive function (Kang & Kim, 2022). And a longitudinal study revealed that frequent binge drinking and marijuana use during the freshman year of college predicted delayed college graduation (White & others, 2017).

When does binge drinking peak during development? As shown in Figure 3, recent data from the *Monitoring the Future* study at the University of Michigan indicate that in 2020 binge drinking peaked at 21 to 26 years of age, with 31 percent in this age group reporting that they had engaged in binge drinking at least once in the last two weeks (Schulenberg & others, 2021).



What kinds of problems are associated with binge drinking in college?
Sean Murphy/Photodisc/Getty Images



How Would You...?

As a social worker, what research-based information would you share with college counselors regarding the prevalence and effects of binge drinking?

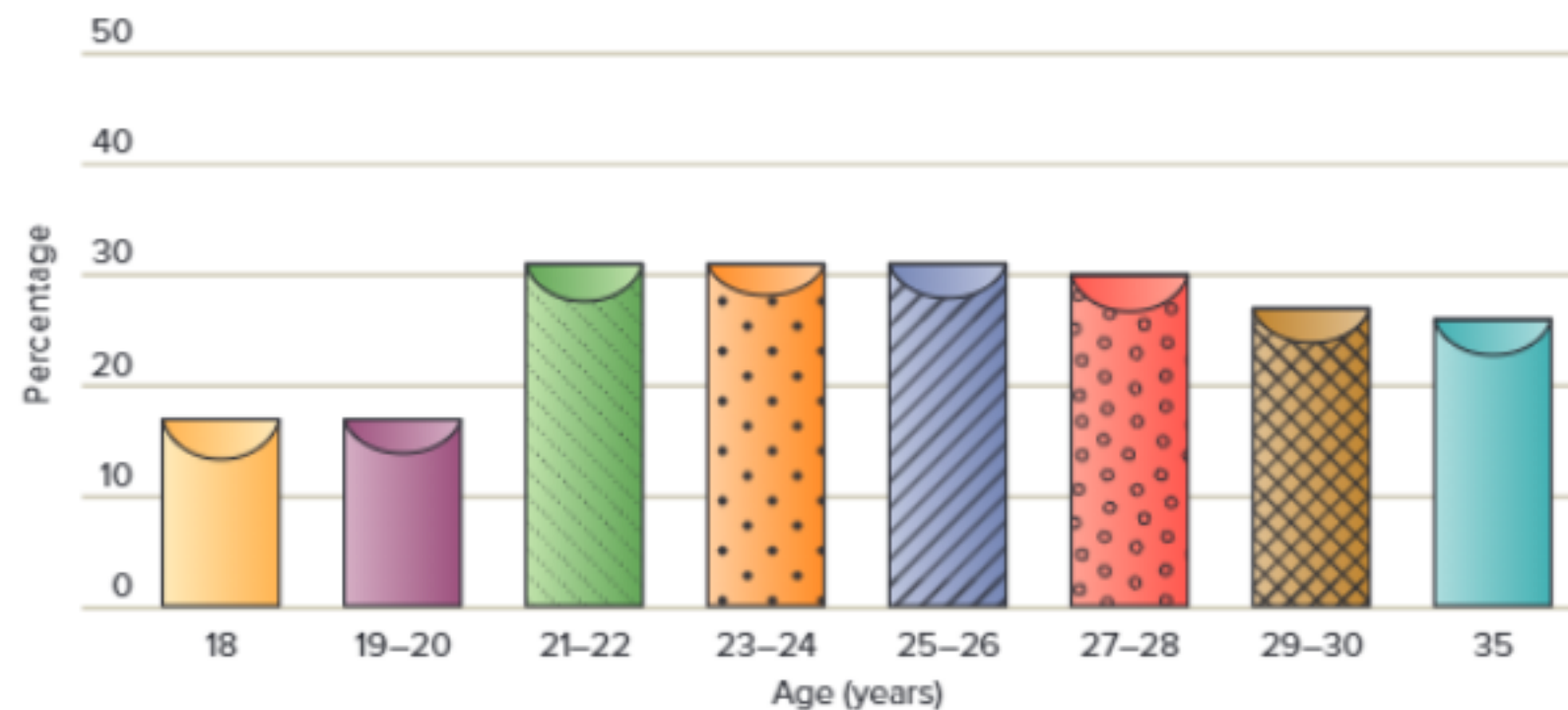


Figure 3 Binge Drinking in the Adolescence–Early Adulthood Transition.

Note that the percentage of individuals engaging in binge drinking peaked at 21 or 22 years of age, remained high through the mid-twenties, then began to decline in the late twenties. Binge drinking was defined as having five or more alcoholic drinks in a row in the previous two weeks.

Source: Schulenberg, J.E., & others (2021)

Sexuality

We have explored how adolescents develop a sexual identity and become sexually active. What happens to their sexuality in adulthood?

Sexual Activity in Emerging Adulthood

At the beginning of emerging adulthood (age 18), surveys indicate that slightly more than 60 percent of individuals have experienced sexual intercourse, but by the end of emerging adulthood (age 25), most individuals have had sexual intercourse (Wesche & Lefkowitz, 2020). Also, the U.S. average age for a first marriage has now climbed to 31 years for men and 28 years for women, higher than at any other point in history (U.S. Census Bureau, 2021). Thus, emerging adulthood is a time during which most individuals are both sexually active and unmarried.

In a recent study of college students, rates of kissing, touching, performing and receiving oral sex, and penetrative sex rates increased between the beginning and end of college (Lefkowitz & others, 2019). Also in this study, contraceptive use increased during the college years, especially for men and for students in serious romantic relationships.

Casual sex is more common in emerging adulthood than it is during the late twenties (Wesche & Lefkowitz, 2020). In a recent research review, overall, individuals rated their casual sexual relationships positively (Wesche, Claxton, & Waterman, 2021). However, they also reported that these casual sexual occurrences were associated with short-term declines in emotional well-being. Women and individuals with less permissive attitudes toward casual sexual encounters experienced worse emotional outcomes, as did individuals who used alcohol prior to casual sexual encounters. Also, not being sexually satisfied and not knowing the sexual partner well were linked to worse emotional outcomes.

A recent trend has involved “hooking up” to have non-relationship sex (from kissing to intercourse) (Hollis & others, 2022). In a recent analysis of first-year college students in six focus groups, a common theme they described was “Sex is easier to get and love is harder to find” (Anders & others, 2020). Also, in a recent study, a greater percentage of emerging adult women reported having sex, hooking up with an acquaintance, using partner characteristics as a reason to hook up, and having negative reactions to their most recent hook-up (Olmstead, Norona, & Anders, 2019). In this study, a greater percentage of emerging adult men reported hooking up with a stranger, meeting at a bar/club, and hooking up at a party. Further, a recent study revealed that both college women and men were more likely to regret hookups with a stranger and when alcohol was involved (Reese-Weber & others, 2022).

In addition to hooking up, another type of casual sex that has increased among emerging adults is “friends with benefits” (FWB), which involves a relationship formed by the integration of friendship and sexual intimacy without an explicit commitment characteristic of an exclusive romantic relationship (Thomas & others, 2023). A recent study found that suicidal ideation was associated with entrance into a friends-with-benefits relationship as well as continuation of the FWB relationship (Dube & others, 2017). Also, a recent study found that individuals participating in a friends-with-benefits relationship reported that sexual exclusivity is an important rule for this type of relationship (van Raalte & others, 2022).

How Would You...?

As a health-care professional, what research-based information might you use to describe casual sex and the increasing rate of casual sex among emerging adults?

Sexual Orientation and Behavior

A national study of sexual behavior in the United States among adults 25 to 44 years of age found that 98 percent of the women and 97 percent of the men said that they



had ever engaged in vaginal intercourse (Chandra & others, 2011). Also in this study, 89 percent of the women and 90 percent of the men reported that they had ever had oral sex with an opposite-sex partner, and 36 percent of the women and 44 percent of the men stated that they had ever had anal sex with an opposite-sex partner.

Also, a national U.S. study found that 30.9 percent of men and 19.1 percent of women 18 to 24 years of age reported being sexually inactive in the past year (Ueda & others, 2020). In this study, the proportions who were sexually inactive from age 25 to 34 (14.1 percent for men, 12.6 percent for women) and from 35 to 44 (8.0 percent and 8.5 percent) were less than for individuals from 18 to 24 and percentage-wise more similar among men and women. Further, sexual inactivity increased among men 18 to 24 years of age and 25 to 34 years of age from 2000 to 2018, and sexual inactivity increased across the same time period for women from 25 to 34 years of age. The increased inactivity for men occurred mainly among unmarried men. Also, men with lower income and men who were working part-time or unemployed were more likely to be sexually inactive, as were women and men who were students. Among the reasons for the decrease in sexual activity may be increased feelings of stress and being overwhelmed when work and intimate relationships compete. In addition, the extensive availability of online entertainment may interfere with sexual activity.

How extensive are gender differences in sexuality? An analysis of almost 8,000 emerging adults found that men had more permissive attitudes, especially about sex in casual relationships, than did women (Sprecher, Treger, & Sakaluk, 2013). Also, a meta-analysis revealed that men reported having slightly more sexual experiences and more permissive attitudes than women for most aspects of sexuality (Petersen & Hyde, 2010). For the following factors, stronger differences were found: Men said that they engaged more often in masturbation, pornography use, and casual sex, and men had more permissive attitudes about casual sex than women did.

Given the media coverage and public attention focusing on the negative aspects of sexuality—such as adolescent pregnancy, sexually transmitted infections, rape, and so on—it is important to underscore that research strongly supports the role of sexuality in well-being (King, 2023). For example, in a Swedish study frequency of sexual intercourse was strongly linked to life satisfaction for both women and men (Brody & Costa, 2009). And in a recent study, sexual activity in adults was linked to experiencing greater well-being the next day (Kashdan & others, 2018). Also in this study, higher reported sexual pleasure and intimacy predicted more positive affect and less negative affect the next day.

Sources of Sexual Orientation

Until the end of the nineteenth century, it was generally believed that people were either heterosexual or homosexual. Today, the more accepted view of sexual orientation depicts it not as an either/or proposition but as a continuum from exclusive male-female relations to exclusive same-sex relations (King, 2023). Some individuals are bisexual, being sexually attracted to people of both sexes.

People sometimes think that bisexuality is simply a stepping stone to same-sex sexuality, while others view it as a sexual orientation in itself or as an indicator of sexual fluidity (King, 2023). Evidence supports the notion that bisexuality is a stable orientation that involves attraction to both sexes (Lippa, 2013).

Compared with men, women are more likely to change their sexual patterns and desires (Knight & Hope, 2012). Women are more likely than men to have sexual experiences with same- and opposite-sex partners, even if they identify themselves strongly as being heterosexual or lesbian (King, 2023). Also, women are more likely than men to identify themselves as bisexual (Gates, 2011).

In the most recent national survey of sexual orientation that included men and women from 18 to 44 years of age, almost three times as many women (17.4 percent) as men (6.2 percent) reported having had same-sex contact (Copen, Chandra, & Febo-Vasquez, 2018). Feelings of attraction only to the opposite sex were more frequent for men (92.1 percent) than for women (81 percent). Also in this study, 92.3 percent of women and 95.1 percent of men described themselves as heterosexual or straight. Further, 1.3 percent of women and 1.9 percent of men said they were homosexual, gay, or lesbian. Also, 5.5 percent of women and 2 percent of men reported that they were bisexual.



What likely determines an individual's sexual preference? (Top) georgeclerk/E+/Getty Images; (middle & bottom) Creatas/Getty Images

Why are some individuals lesbian, gay, or bisexual (LGB) while others are heterosexual? Speculation surrounding this question has been extensive (Hyde & De Lamater, 2024). All people, regardless of their sexual orientation, have similar physiological responses during sexual arousal and seem to be aroused by the same types of tactile stimulation. Investigators typically find no differences between sexual minority individuals and heterosexuals in a wide range of attitudes, behaviors, and adjustments (Fingerhut & Peplau, 2013).

A special concern involving sexual minority individuals are the hate crimes and stigma-related experiences they encounter (Hoy-Ellis, 2023). In a recent study, sexual minority reports of past year general discrimination, victimization, and health-care discrimination peaked in early adulthood and middle adulthood (Rice & others, 2021).

Recently, researchers have explored the possible biological basis of same-sex relations. The results of hormone studies have been inconsistent. If gay men are given male sex hormones (androgens), their sexual orientation doesn't change—their sexual desire merely increases. A very early prenatal critical period might influence sexual orientation (Swift-Gallant & others, 2022). If this critical-period hypothesis turns out to be correct, it would explain why clinicians have found that sexual orientation is difficult, if not impossible, to modify.

Researchers have also examined genetic influences on sexual orientation by studying twins. A Swedish study of almost 4,000 twins found that only about 35 percent of the variation in homosexual behavior in men and 19 percent in women could be explained by genetic differences (Langstrom & others, 2010). This result suggests that although genes likely play a role in sexual orientation, they are not the only factor involved (King, 2023).

An individual's sexual orientation—same-sex, heterosexual, or bisexual—is most likely determined by a combination of genetic, hormonal, cognitive, and environmental factors (King, 2023). Most experts on same-sex relations believe that no one factor alone causes sexual orientation and that the relative weight of each factor can vary from one individual to the next. That said, it has become clear that no matter whether people are heterosexual, gay, lesbian, or bisexual, they cannot be talked out of their sexual orientation (King, 2023).

How Would You...?

As a **clinical psychologist**, what research-based information would you give in a workshop for college students on exploring sexual orientations?



Attitudes and Behavior of Lesbians and Gay Men

Many gender differences that appear in heterosexual relationships occur in same-sex relationships (Savin-Williams, 2019). For example, lesbians have fewer sexual partners than gays, and lesbians have less permissive attitudes about casual sex outside a primary relationship than gays do (Fingerhut & Peplau, 2013).

Sexually Transmitted Infections

Sexually transmitted infections (STIs) are diseases that are primarily contracted through sexual relations—intercourse as well as oral-genital and anal-genital sex. STIs affect about one of every six U.S. adults (National Center for Health Statistics, 2023). Among the most prevalent STIs are bacterial infections—such as gonorrhea, syphilis, and chlamydia—and STIs caused by viruses—such as AIDS (acquired immune deficiency syndrome), genital herpes, and genital warts. Figure 4 describes these sexually transmitted infections.

sexually transmitted infections (STIs) Diseases that are contracted primarily through sexual contact.

STI	Description/cause	Incidence	Treatment
Gonorrhea	Commonly called the “drip” or “clap.” Caused by the bacterium <i>Neisseria gonorrhoeae</i> . Spread by contact between infected moist membranes (genital, oral-genital, or anal-genital) of two individuals. Characterized by discharge from penis or vagina and painful urination. Can lead to infertility.	500,000 cases annually in U.S.	Penicillin, other antibiotics
Syphilis	Caused by the bacterium <i>Treponema pallidum</i> . Characterized by the appearance of a sore where syphilis entered the body. The sore can be on the external genitals, vagina, or anus. Later, a skin rash breaks out on palms of hands and bottom of feet. If not treated, can eventually lead to paralysis or even death.	100,000 cases annually in U.S.	Penicillin
Chlamydia	A common STI named for the bacterium <i>Chlamydia trachomatis</i> , an organism that spreads by sexual contact and infects the genital organs of both sexes. A special concern is that females with chlamydia may become infertile. It is recommended that adolescent and young adult females have an annual screening for this STI.	About 3 million people in U.S. annually	Antibiotics
Genital herpes	Caused by a family of viruses with different strains. Involves an eruption of sores and blisters. Spread by sexual contact.	One of five U.S. adults	No known cure but antiviral medications can shorten outbreaks
AIDS	Caused by a virus, the human immunodeficiency virus (HIV), which destroys the body’s immune system. Semen and blood are the main vehicles of transmission. Common symptoms include fevers, night sweats, weight loss, chronic fatigue, and swollen lymph nodes.	More than 300,000 cumulative cases of HIV virus in U.S. 25–34-year-olds; epidemic incidence in sub-Saharan countries	New treatments have slowed the progression from HIV to AIDS; no cure
Genital warts	Caused by the human papillomavirus, which does not always produce symptoms. Usually appear as small, hard painless bumps in the vaginal area, or around the anus. Very contagious. Certain high-risk types of this virus cause cervical cancer and other genital cancers. May recur despite treatment. A new HPV preventive vaccine, Gardasil, has been approved for girls and women 9–26 years of age.	About 5.5 million new cases annually; considered the most common STI in the U.S.	A topical drug, freezing, or surgery

Figure 4 Sexually Transmitted Infections.

No single disease has had a greater impact on sexual behavior, or created more public fear in the last several decades, than infection with the human immunodeficiency virus (HIV) (Hyde & DeLamater, 2024). HIV is a virus that destroys the body’s immune system. Once a person is infected with HIV, the virus breaks down and overpowers the immune system, which leads to AIDS. An individual sick with AIDS has such a weakened immune system that a common cold can be life-threatening.

In 2019, approximately 1.2 million people in the United States were living with an HIV infection (HIVgov, 2022). Also in 2019, approximately 34,800 new HIV infections occurred in the United States (HIVgov, 2022). Male-male sexual contact continued to be the most frequent AIDS transmission category. Gay and bisexual men accounted for 69 percent of all new HIV diagnoses (HIVgov, 2022). The HIV rate was highest for the 25 to 34 age group.

New HIV infections in the United States have been reduced by more than two-thirds since the height of the epidemic in the mid-1980s. Also, there is an increasing effort to reduce HIV infections through pre exposure prophylaxis (PrEP) awareness, which involves taking a daily pill that can reduce the probability of getting HIV by as much as 90 percent (Moskowitz & others, 2021).

Globally, the total number of individuals living with HIV was 38.4 million in 2021, with the majority of these individuals with HIV living in sub-Saharan Africa (UNAIDS, 2022). Worldwide there were 1.5 million new cases of HIV in 2021, a reduction of 52 percent since the peak in 1997 (UNAIDS, 2022). Currently, only about 85 percent of individuals with HIV know they have the disease (UNAIDS, 2022). Approximately half of all new HIV infections around the world occur in the 15- to 24-year-old age

category. In 2021, 54 percent of all people living with HIV were girls and women (UNAIDS, 2022). The good news is that global rates of HIV infection have decreased considerably in recent years (UNAIDS, 2022).

What are some good strategies for protecting against HIV and other sexually transmitted infections? They include the following:

- *Knowing your own and your partner's risk status.* Anyone who has had previous sexual activity with another person might have contracted an STI without being aware of it. Spend time getting to know a prospective partner before you have sex. Use this time to inform the other person of your STI status and inquire about your partner's. Remember that many people lie about their STI status.
- *Obtaining medical examinations.* Many experts recommend that couples who want to begin a sexual relationship have a medical checkup to rule out STIs before engaging in sex. If cost is an issue, contact your campus health service or a public health clinic.
- *Having protected, not unprotected, sex.* When correctly used, latex condoms help to prevent many STIs from being transmitted. Condoms are most effective in preventing gonorrhea, syphilis, chlamydia, and HIV. They are less effective against the spread of herpes.
- *Not having sex with multiple partners.* One of the best predictors of getting an STI is having sex with multiple partners. Having more than one sex partner elevates the likelihood of encountering an infected partner.

Cognitive Development

Are there changes in cognitive performance during these years? To explore the nature of cognition in early adulthood, we focus on issues related to cognitive stages and creative thinking.

Cognitive Stages

Are young adults more advanced in their thinking than adolescents are? Let's examine how Piaget and others have answered this intriguing question.

Piaget's View

Piaget concluded that an adolescent and an adult think qualitatively in the same way. That is, Piaget argued that at approximately 11 to 15 years of age, adolescents enter the formal operational stage, which is characterized by more logical, abstract, and idealistic thinking than the concrete operational thinking of 7- to 11-year-olds. Piaget did believe that young adults are more quantitatively advanced in their thinking in the sense that they have more knowledge than adolescents possess. He also believed, as do information-processing psychologists, that adults especially increase their knowledge in a specific area, such as a physicist's understanding of physics or a financial analyst's knowledge about finance. According to Piaget, however, formal operational thought is the final stage in cognitive development, and it characterizes adults as well as adolescents.

Some developmentalists theorize it is not until adulthood that many individuals consolidate their formal operational thinking. That is, they may begin to plan and hypothesize about intellectual problems in adolescence, but they become more systematic and sophisticated in their thinking as young adults. Nonetheless, even many adults do not think in formal operational ways (Kuhn, 2009).

Postformal Thought

It has been proposed that the idealism of Piaget's formal operational stage declines in young adults and is replaced by more realistic, pragmatic thinking. It also has been

proposed that young adults move into a new qualitative stage of cognitive development called postformal thought (Sinnott, 2003). **Postformal thought** is:

postformal thought Thinking that is reflective, relativistic, and contextual; provisional; realistic; and influenced by emotions.

- *Reflective, relativistic, and contextual.* As young adults engage in solving problems, they might think deeply about many aspects of work, politics, relationships, and other areas of life (Labouvie-Vief, 1986). They find that what might be the best solution to a problem at work (with a boss or co-worker) might not be the best solution at home (with a romantic partner). Thus, postformal thought holds that the correct answer to a problem requires reflective thinking and may vary from one situation to another. Some psychologists argue that reflective thinking continues to increase and becomes more internal and less contextual in middle age (Labouvie-Vief, Gruhn, & Studer, 2010; Mascolo & Fischer, 2010).
- *Provisional.* Many young adults also become more skeptical about the truth and seem unwilling to accept an answer as final. Thus, they come to see the search for truth as an ongoing and perhaps never-ending process.
- *Realistic.* Young adults understand that thinking can't always be abstract. In many instances, it must be realistic and pragmatic.
- *Recognized as being influenced by emotion.* Emerging and young adults are more likely than adolescents to understand that their thinking is influenced by emotions (Girgis & others, 2018; Labouvie-Vief, 2009). However, too often negative emotions produce thinking that is distorted and self-serving at this point in development.

In addition to the characteristics just described for a possible fifth, postformal stage, a recent study explored wisdom and meaning as important developments in emerging adulthood (Webster & others, 2018). This study found that the search for and presence of meaning was linked to wisdom, which was assessed with five components: critical life experiences, reminiscence/reflectiveness, openness to experience, emotional regulation, and humor. We will further explore meaning in life in the chapter on "Socioemotional Development in Middle Adulthood" and wisdom in the chapter on "Cognitive Development in Late Adulthood."



What are some ways that young adults might think differently from adolescents?
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How Would You...?

As an **educator**, how would you explain the characteristics of postformal thought to other educators who are seeking to encourage cognitive development in their students?



Creativity

Early adulthood is a time of great creativity for some people. Researchers have found that creativity often peaks in the forties, then declines (Simonton, 2021). However, any generalization about a link between age and creative accomplishments must be qualified by consideration of (1) the size of the decline and (2) differences across domains (Baer, 2022).

Any consideration of decline in creativity with age must take into account the field of creativity involved (Kandler & others, 2016). In fields such as philosophy and history, older adults often show as much creativity as they did when they were in their thirties and forties. By contrast, in fields such as lyric poetry, abstract math, and theoretical physics, the peak of creativity is often reached in the twenties or thirties.

Researchers have found that personality traits are linked to creativity (Feist, 2018). In one recent study, the personality trait of openness to experience predicted creativity in the arts, while intellectual capacity predicted creativity in the sciences (Kaufman & others, 2016).

Can you make yourself more creative? Mihaly Csikszentmihalyi (1995) interviewed 90 leading figures in art, business, government, education, and science to learn how



Mihaly
Csikszentmihalyi,
in the setting where
he gets his most
creative ideas.
*When and where do
you get your most
creative thoughts?*
Dr. Mihaly
Csikszentmihalyi

creativity works. He discovered that creative people regularly experience a state he calls *flow*, a heightened state of pleasure experienced when we are engaged in mental and physical challenges that absorb us. Csikszentmihalyi (2000) believes everyone is capable of achieving flow. Based on his interviews with some of the most creative people in the world, the first step toward a more creative life is cultivating your curiosity and interest. How can you do this?

- *Try to be surprised by something every day.* It may be something you see, hear, or read about. Become absorbed in a lecture or a book. Be open to what the world is telling you. Life is a stream of experiences. Swim widely and deeply in it, and your life will be richer.
- *Try to surprise at least one person every day.* In a lot of things you do, you have to be predictable and patterned. Do something different for a change. Ask a question you normally would not ask. Invite someone to go to a show you haven't seen or a museum you never have visited.
- *Write down each day what surprised you and how you surprised others.* Most creative people keep a diary, notes, or lab records to ensure that their experience is not fleeting or forgotten. Start with a specific task. Each evening, record the most surprising event that occurred that day and your most surprising action. After a few days, reread your notes and reflect on your past experiences. After a few weeks, you might see a pattern of interest emerging in your notes, one that might suggest an area you can explore in greater depth.
- *When something sparks your interest, follow it.* Usually when something captures your attention, it is short-lived—an idea, a song, a flower. Too often we are too busy to explore the idea, song, or flower further. Or we think these areas are none of our business because we are not experts about them. Yet the world is our business. We can't know which part of it is best suited to our interests until we make a serious effort to learn as much about as many aspects of it as possible.
- *Wake up in the morning with a specific goal to look forward to.* Creative people wake up eager to start the day. Why? Not necessarily because they are cheerful, enthusiastic types but because they know that there is something meaningful to accomplish each day, and they can't wait to get started.
- *Spend time in settings that stimulate your creativity.* In Csikszentmihalyi's (1995) research, he gave people an electronic pager and beeped them randomly at different times of the day. When he asked them how they felt, they reported the highest levels of creativity when walking, driving, or swimming. I (your author) do my most creative thinking when I'm jogging. These activities are semiautomatic in that they take a certain amount of attention while leaving some time free to make connections among ideas. Another setting in which highly creative people report coming up with novel ideas is the sort of half-asleep, half-awake state we are in when we are deeply relaxed or barely awake.

Achievement, Careers, and Work

Achieving success; choosing, establishing, and developing a career; and working are important themes of early adulthood. What are some of the factors that go into achieving success in life, having a rewarding career, and working effectively?

Achievement

In "Cognitive Development in Early Childhood" and "Socioemotional Development in Middle and Late Childhood," we described some important aspects of achievement in children's lives that continue to be important in adults' achievement. These include *self-efficacy*, the belief that one can master a situation and produce favorable outcomes

(Bandura, 2012) and *mindset*, the cognitive view individuals develop for themselves (Dweck, 2022). Recall that a *growth mindset*, in which people believe their qualities can change and improve through effort, is linked to success and achievement, whereas a *fixed mindset*, in which they believe their qualities are carved in stone and cannot change, is associated with lower achievement and less success. In a recent study of adults, engaging in a strategic mindset was critical in achieving goals (Chen & others, 2020). This mindset involves asking oneself strategy-eliciting questions such as “What can I do to help myself?” and “Is there a way to do this even better?” in the face of challenges or inadequate progress. In this study, those with a strategic mindset used more metacognitive strategies and in turn had higher college grade point averages and showed greater progress toward their professional, educational, and health goals. Also remember our discussion of *delay of gratification*—engaging in self-control by waiting until later to obtain something more valuable rather than immediately seeking satisfaction with something less valuable (Mischel, 2014). Here we explore other factors that are important in adults’ success and achievement: motivation; goal-setting, planning, and self-monitoring; and grit.

extrinsic motivation Doing something to obtain something else (the activity is a means to an end).

intrinsic motivation Doing something for its own sake; involves factors such as self-determination and opportunities to make choices.

Motivation

In thinking about a person’s motivation, psychologists have examined whether it is advantageous to be extrinsically or intrinsically motivated (Stockkamp & others, 2023). **Extrinsic motivation** involves doing something to obtain something else (the activity is a means to an end). Extrinsic motivation is often influenced by external incentives such as rewards and punishments. For example, an individual might work very hard at a job to make money. **Intrinsic motivation** involves the internal motivation to do something for its own sake (the activity is an end in itself). For example, an individual might work very hard at a job because they are self-motivated.

An overwhelming conclusion of motivation research is that it is important to be intrinsically motivated (Miraj & others, 2022). That said, the real world includes both intrinsic and extrinsic motivation, and too often intrinsic and extrinsic motivation have been pitted against each other as polar opposites. In many aspects of students’ lives, both intrinsic and extrinsic motivation are at work (Schunk, 2020). Further, intrinsic and extrinsic motivation can operate simultaneously. Thus, people might work hard at a job because they enjoy the work and like learning about it (intrinsic), and because they want to earn a good salary (extrinsic). Keep in mind, though, that many psychologists suggest that extrinsic motivation by itself is not a good strategy.

How Would You...?

As an **educator**, how would you explain the concepts of intrinsic and extrinsic motivation when discussing the use of extra credit in the classroom?



Goal-Setting, Planning, and Self-Monitoring

Setting goals, planning how to reach those goals, and engaging in self-regulation and monitoring of progress toward goals are important aspects of achievement (Ehrlich, 2023). Individuals should set both long-term and short-term goals. A long-term goal might be “I want to get an undergraduate degree in four years.” A short-term goal might be “I want to get an A on the biology test on Monday.”

Another good strategy is to set challenging goals. A challenging goal is a commitment to self-improvement. Strong interest and involvement in activities are sparked by challenges. Goals that are easy to reach generate little interest or effort. However, goals should be optimally matched to the individual’s skill level. If goals are unrealistically high, the result will be repeated failures that lower the individual’s self-efficacy.

It is not enough just to set goals. In order to achieve them, it also is important to plan how to reach those goals. Being a good planner means managing time effectively, setting priorities, and being organized. Individuals should not only plan their next week’s



How are goals, planning, and self-monitoring involved in adolescent achievement?
Westend61/Getty Images



Meredith MacGregor (pictured as a senior at Fairview High School in Boulder, Colorado) was one of Colorado's top high school long-distance runners. An aspiring scientist, she was awarded the Intel Young Scientist Award. After obtaining advanced degrees in astronomy and astrophysics from Harvard University, followed by a post-doctoral degree from Carnegie Mellon University, she is now a professor at the Center for Astrophysics and Space Astronomy at the University of Colorado–Boulder. *What are some aspects of grit that likely characterize Dr. MacGregor?*

Kevin Moloney

activities but also monitor how well they are sticking to their plan. Once engaged in a task, they need to monitor their progress, judge how well they are doing on the task, and evaluate the outcome to regulate what they will do in the future.

grit Involves passion and persistence in achieving long-term goals.

Grit

Recently, in the continuing effort to study key aspects of achievement, researchers have explored the concept of **grit**, which involves passion and persistence in achieving long-term goals (Lee, Reasoner, & Lee, 2023). Research indicates that grit is linked to academic engagement and positive results, such as obtaining a higher grade point average (Hossain & others, 2022). For example, a recent study revealed that college students with a higher level of grit received better grades on their final exams during the COVID-19 pandemic (Sulla & others, 2022). In another study, nursing students with more grit had less academic burnout and lower levels of job-seeking stress (Kang & Choi, 2022). Further, recent research revealed that during the COVID-19 pandemic college students with more grit reported having better eating and sleeping behaviors than students who had less grit (Martin & others, 2023). Recent research also suggests that life purpose commitment (Hill & others, 2016), hope (Ekinici & Koc, 2023), mindfulness (Lee, 2022), and goal commitment (Tang & others, 2021) are linked to grit.

Among the questions to ask yourself to determine if you have grit are the following (Clark & Malecki, 2019):

- Do I push myself to be my best?
- Do I work hard to reach goals, no matter how long it takes to reach them?
- Once I set a goal, do I strive to overcome any challenges I might encounter?
- How passionate am I about the work I am doing?

Now that we have explored some key aspects of achieving success, let's turn our attention to developmental changes in careers and then examine various aspects of the world of work.

Careers

What are some developmental changes young adults experience as they choose a career? How effectively are individuals finding a path to purpose today?

Developmental Changes in Careers

Many children have idealistic fantasies about what they want to be when they grow up. For example, many young children want to be superheroes, sports stars, or movie stars. As high school students, they often begin to think about careers in a somewhat less idealistic way. In their late teens and early twenties, their career decision making has usually turned more serious as they explore different career possibilities and zero in on the career they want to enter. In college, this often means choosing a major or specialization that is designed to prepare them for a career in a particular field. By their early and mid-twenties, many individuals have completed their education or training and entered a full-time occupation. From the mid-twenties through the remainder of early

How Would You...?

As a **social worker**, what important characteristics of goal-setting and grit might you emphasize to motivate your clients to achieve their goals?



adulthood, individuals often seek to establish their emerging career in a particular field. They may work hard to move up the career ladder and improve their financial standing.

Phyllis Moen (2009a) described the career mystique, which includes ingrained cultural beliefs that engaging in hard work for long hours through adulthood will produce a path to status, security, and happiness. Many individuals envision achieving the American dream of upward mobility by climbing occupational ladders. However, the lockstep career mystique has never been a reality for many workers, especially ethnic minority individuals, women, and poorly educated adults. Further, the career mystique has increasingly become a myth for many individuals in middle-income occupations as global outsourcing of jobs and widespread layoffs during the coronavirus pandemic have led to reduced job security for millions of Americans.

Finding a Path to Purpose

In his book *The Path to Purpose: Helping Our Children Find Their Calling in Life*, William Damon (2008) suggested that purpose is a missing ingredient in many adolescents' and emerging adults' achievement and career development. Too many youth drift aimlessly through their high school and college years, Damon says, engaging in behavior that places them at risk for not fulfilling their potential and not finding a life pursuit that energizes them.

In interviews with 12- to 22-year-olds, Damon found that only about 20 percent had a clear vision of where they wanted to go in life, what they wanted to achieve, and why. A large percentage—about 60 percent—had engaged in some potentially purposeful activities, such as service learning or fruitful discussions with a career counselor—but they still did not have a real commitment or any reasonable plans for reaching their goals. And slightly more than 20 percent expressed no aspirations and, in some instances, said they didn't see any reason to have aspirations.

Damon concludes that most teachers and parents communicate the importance of studying hard and getting good grades but rarely discuss where these accomplishments might lead young adults. Damon emphasizes that too often students focus only on short-term goals and don't explore the big, long-term picture of what they want to do with their life (Damon & Malin, 2020). The following interview questions that Damon (2008, p. 135) has used in his research are good springboards for helping individuals to reflect on their purpose:

- What's most important to you in your life?
- Why do you care about those things?
- Do you have any long-term goals?
- Why are these goals important to you?
- What does it mean to have a good life?
- What does it mean to be a good person?
- If you were looking back on your life now, how would you like to be remembered?

Recent research has provided support for the importance of purpose in people's lives (Chen & Cheng, 2020; Damon & Malin, 2020). In one study, purpose predicted



Hari Prabhakar (*in rear*) at a screening camp in India that he created as part of his Tribal India Health Foundation. Hari reflects William Damon's concept of finding a path to purpose. His ambition is to become an international health expert. A 2006 graduate from Johns Hopkins University (with a double major in public health and writing and a 3.9 GPA), he pursued many activities outside the classroom, in the health field. Hari created the Tribal India Health Foundation, which provides assistance in bringing low-cost health care to rural areas in India. Juggling roles as a student and as the foundation's director, Hari spent 15 hours a week leading Tribal India Health during his undergraduate years. Hari said, "I have found it very challenging to coordinate the international operation. . . . It takes a lot of work, and there's not a lot of free time. But it's worth it when I visit our patients and see how they and the community are getting better" (Johns Hopkins University, 2006).

Hari Prabhakar

emerging adults' well-being (Hill & others, 2016). In another study, a high sense of purpose in life was associated with a lower incidence of cardiovascular disease and a longer life (Cohen, Bavishi, & Rozanski, 2016).

Work

In this final section, we'll examine how work affects people's lives, the role of work in college, the occupational outlook, unemployment, dual-earner couples, and diversity in the workplace.

The Impact of Work

Work defines people in fundamental ways (Nickels & others, 2024). It is an important influence on their financial standing, housing, the way they spend their time, where they live, their friendships, and their health (Floyd & Cardon, 2024). Some people define their identity through their work. Work also creates a structure and rhythm to life that is often missed when individuals do not work for an extended period. When they are unable to work, many individuals experience emotional distress and reduced self-esteem.

Most individuals spend about one-third of their lives at work. In one survey, U.S. individuals 18+ years old who were employed full-time worked an average of 47 hours per week, almost a full workday longer than the standard 9-to-5 five days a week schedule (Saad, 2014). In this survey, half of all individuals working full-time reported that they worked more than 40 hours a week, and nearly 40 percent said they worked 50 hours a week or more. Only 8 percent said they worked less than 40 hours per week.

At the beginning of 2020, the job outlook for college graduates in the United States was very favorable, with employers planning to increase college hiring by 5.8 percent over 2019 hiring (NACE, 2020). However, as in many other aspects of life, the outlook for college graduate employment declined steeply in March and April of 2020 with the onset of the coronavirus pandemic. However, many employers planned to hire significantly more college graduates from the class of 2021, expecting to hire 7.2 percent more graduates than they did in 2020 (NACE, 2021). In the spring of 2022, colleges planned to hire 31.6 percent more new graduates from the class of 2022 than they had hired from the class of 2021 (NACE, 2022). In April 2023, employers projected hiring 4 percent more new college graduates than they did from the class of 2022, down from the projection employers made in the fall of 2022 of hiring 14 percent more graduates (NACE, 2023).

A trend in the U.S. workforce is the disappearing long-term career for an increasing number of adults, especially men in private-sector jobs. Among the reasons for the reduced number of long-term jobs is the dramatic increase in technology and cheaper labor in other countries.

Many young and older adults are working at a series of jobs, and many work in short-term jobs. Early careers are especially unstable as some young workers move from "survival jobs" to "career jobs" in the process of finding a job that matches their personal interests and goals (Staff, Mont'Alvao, & Mortimer, 2015).

An important consideration regarding work is how stressful it is (Hellyer, 2023). One study revealed that stressors at work were linked to arterial hypertension in employees (Lamy & others, 2014). A recent study indicated that increases in job strain increased workers' insomnia, while decreases in job strain reduced their insomnia (Halonen & others, 2017). Further, in a recent study work-related stress and night shift work increased workers' obesity (da Silva & others, 2021). And a recent European study of 15 countries found that from 1995 to 2015 work stress increased, especially in lower-skilled occupations (Rigo & others, 2021).

Responding to global competition and pressure from financial markets, many companies are asking employees to do more with less, even as new technologies normalize 24/7 job expectations. In *Overload: How Good Jobs Went Bad And What We Can Do About It*, Erin Kelly and Phyllis Moen (2020) document how intensification of work has created chronic stress for individuals, producing burnout, attrition, and poor job

performance. Their research includes hundreds of interviews with employees and managers. They found that when companies adopt creative and practical work redesigns that give workers more control over how and where they work and when they encourage managers to evaluate performance in new ways, the companies see improvements in employees' health, well-being, and ability to manage their personal and work lives. Also, when these changes are implemented, companies benefit from increased job satisfaction and lower turnover. Such changes can—and should—be made on a wide scale in companies.

The contexts in which people work in the United States and around the world changed dramatically in 2020 when the coronavirus pandemic forced many employers to require their employees to work at home to comply with social distancing and quarantine requirements in their communities. As the virus subsides, it is possible that greater numbers of employers may offer their employees a choice of working at home or at the employer's office, or allow them to engage in a hybrid schedule that involves working at home some days and at the office on other days.

Work During College

The percentage of full-time undergraduate students who were employed in 2020 (40 percent) was lower than in 2015 (43 percent) (National Center for Education Statistics, 2022). Similarly, the percentage of part-time undergraduates who were employed was lower in 2020 (74 percent) than in 2015 (78 percent).

Holding down a job during college can pay for schooling or help offset some of its costs, but working also can restrict students' opportunities to learn. For those who identified themselves primarily as students, one national study found that as the number of hours worked per week increased, their grades suffered (National Center for Education Statistics, 2002). Thus, college students need to carefully examine whether the number of hours they work is having a negative impact on their college success.

How Would You...?

As an educator, what research-based information would you share with a college student regarding the impact of employment on grades?



Monitoring the Occupational Outlook

As you explore the type of work you are likely to enjoy and in which you can succeed, it is important to become knowledgeable about different fields and companies. Occupations may have many job openings one year but few in another year as economic conditions change. Thus, it is critical to keep up with the occupational outlook in various fields.

An excellent resource for doing this is the U.S. government's *Occupational Outlook Handbook*, which is revised every two years. According to the 2022–2023 edition, nurse practitioners, wind turbine service technicians, ushers, lobby attendants, ticket takers, motion picture projectionists, restaurant cooks, data scientists, athletes, sports competitors, information security analysts, and statisticians are the job categories that are projected to grow most rapidly through 2031 (Occupational Outlook Handbook, 2022–2023). Most of the highest-paying occupations require a college degree.

College career counselors and academic advisors are trained to provide guidance and support regarding future career opportunities, the required coursework, and how to stay on track to meet academic and career goals. See the *Careers in Life-Span Development* profile to learn how one advisor and her school's programs are helping students to achieve success.

Unemployment

A major shift in unemployment occurred at the onset of the coronavirus in 2020. In 2019 and the first two months of 2020, the U.S. unemployment rate was at its lowest point in decades. Then it abruptly took a steep upturn in March 2020 as companies shut their doors temporarily or permanently and let go of employees. In April and May of 2020,

Careers in life-span development

Anna Boyer-Chadwick, Academic Advisor

Anna Boyer-Chadwick is an academic advisor at the University of Texas–San Antonio. After graduating from Central Washington University with a bachelor's degree in psychology, she completed a master's degree in educational leadership at the University of Nevada–Las Vegas. Prior to becoming an academic advisor, Boyer-Chadwick worked in several different areas in higher education, including housing/residence life, orientation, and student conduct (UTSA Pride Faculty & Staff Association, 2023).

At UTSA, each student is assigned an academic advisor who helps them to develop meaningful, personalized pathways toward academic success, to stay on track, and to achieve their ultimate career goals. UTSA's Resilience and Retention (R&R) Advising Program, which reaches out to help students who are academically dismissed or denied admission into the major of their choice and are at a high risk of dropping out of college, has seen tremendous success in the form of increased retention and graduation rates. Participating students sign a success agreement, commit to utilizing at least two academic support resources, and meet monthly with R&R advisors to discuss finances, career knowledge, and



Anna Boyer-Chadwick (left) counsels a college student.
Anna Boyer-Chadwick

skills that can be gained from any degree. The program has become a nationally recognized model for Latino student success, winning a number of awards including the Outstanding Advising Program Merit Award from NACADA: The Global Community for Academic Advising. Boyer-Chadwick has also been recognized by NACADA for her work as an advisor.

the unemployment rate decreased as the economy began gradually reopening but remained much higher (13.3 percent in May) than it had been in the early months of 2020 (3.5 percent in February). However, in October 2021, the unemployment rate had dropped to 4.6 percent, and in April 2023, it had dropped even further to 3.4 percent.

Unemployment produces stress regardless of whether the job loss is temporary, cyclical, or permanent (Offenberg & others, 2022). Researchers have linked unemployment to physical problems (such as heart attack and stroke), emotional problems (such as depression and anxiety), marital difficulties, substance abuse, and homicide (Hagen & others, 2022). Further, in a recent Finnish study, alcohol-related death rates continued to be higher for individuals up to 20 years after they were unemployed (Junna, Moustgaard, & Martikainen, 2021). And a recent study found that following a period of unemployment, recovery of a sense of well-being upon reemployment was fast and enduring even when individuals took less favorable employment upon returning to work (Zhou & others, 2019).

Dual-Earner Couples

Dual-earner couples may have special problems finding a balance between work and the rest of life (Crafford & Koekemoer, 2022). If both partners are working, who cleans the house or calls the repairman or takes care of the other endless details involved in maintaining a home? If the couple has children, who is responsible for being sure that the children get to school or to piano lessons on time, and who writes the notes to approve field trips or meets with the teacher or makes the dental appointments?

Many dual-earner couples engage in a range of adaptive strategies to coordinate their work and manage the family side of the work-family equation. Researchers have found that even though couples may strive for gender equality in dual-earner families, gender inequalities persist. For example, women still do not earn as much as men in the same jobs, and this inequity contributes to gender divisions in how much time each

partner spends in paid work, homemaking, and caring for children. Thus, the decisions made by dual-earner couples often favor men's greater earning power and cause women to spend more time than men in homemaking and caring for children (Moen, 2009b). One study indicated that women reported more family interference from work than did men (Allen & Finkelstein, 2014). Another study found that partner coping, having a positive attitude toward multiple roles, using planning and management skills, and not having to cut back on professional responsibilities were linked to better relationships between dual earners (Matias & Fontaine, 2015).

Summary

The Transition from Adolescence to Adulthood

- Emerging adulthood, the time of transition from adolescence to adulthood, is characterized by experimentation and exploration. Today's emerging and young adults are experiencing emerging and early adulthood quite differently from their counterparts in earlier generations.
- The transition from high school to college can involve both positive and negative features.

Physical Development

- Peak physical performance is often reached between 19 and 26 years of age. Then, toward the latter part of early adulthood, a detectable slowdown in physical performance is apparent for most individuals.
- Health problems in emerging and young adults may include obesity, a serious problem throughout the United States. Binge drinking is a special problem among U.S. college students, but by the mid-twenties alcohol and drug use often decreases.

Sexuality

- Patterns of sexual activity change during emerging adulthood.
- An individual's sexual orientation likely stems from a combination of genetic, hormonal, cognitive, and environmental factors.
- Sexually transmitted infections, also called STIs, are contracted primarily through sexual contact.

Cognitive Development

- It has been proposed that the idealism of Piaget's formal operational stage declines in young adults and is replaced

by more realistic, pragmatic thinking. A qualitatively different, fifth cognitive stage called postformal thought also has been proposed.

- Creativity peaks in adulthood, often during the forties, and then declines. Csikszentmihalyi proposed that the first step toward living a creative life is to cultivate curiosity and interest.

Achievement, Careers, and Work

- Among the key factors in achieving success are being highly motivated, especially in terms of intrinsic motivation, setting goals, planning, and self-monitoring, and having grit.
- Thoughts about career choice for adolescents and young adults reflect developmental changes. Damon argues that too many individuals have difficulty finding a path to purpose today.
- Work defines people in fundamental ways and is a key aspect of their identity. Working during college can have a positive outcome, but it may also have a negative impact on grades. Jobs that require a college education will be the fastest-growing and highest-paying careers in the United States during the next decade. Unemployment has a number of negative outcomes for individuals. As the number of women working outside the home has increased, new issues involving work and family have arisen.

Key Terms

aerobic exercise
emerging adulthood

extrinsic motivation
grit

intrinsic motivation
postformal thought

sexually transmitted infections
(STIs)